

PROCEEDING OF

2nd INTERNATIONAL

CONFERENCE ON

CLOUD COMPUTING & DATA

SCIENCE

Organized By

PG DEPARTMENT OF COMPUTER SCIENCE AND
APPLICATIONS

&

INTERNAL QUALITY ASSURANCE CELL

13th October 2025

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ICCCDS' 25

2nd INTERNATIONAL CONFERENCE ON CLOUD COMPUTING & DATA SCIENCE

13th October 2025

PROCEEDING

PERI

COLLEGE OF ARTS AND SCIENCE

**PG DEPARTMENT OF COMPUTER SCIENCE AND
APPLICATIONS
&
INTERNAL QUALITY ASSURANCE CELL**

PERI ICCCDs' 2025

ABOUT PERI

PERI College of Arts and Science is established by the PAPPATHI AMMAAL Educational Trust with the objective of providing quality technical education. The College is built at Mannivakkam, Chennai in a sprawling area replete with well equipped labs, A/C transport facility, modern laboratories, cafeteria and a library with latest technology.

PERI College of Arts and Science offers UG Courses B.Com General, B.Com Accounting & Finance, B.Sc Chemistry, B.Sc Computer Science, B.Sc. Visual Communication , B.Sc Microbiology, Bio-Technology, B.Sc Psychology, B.C.A Computer Applications, B.A Tamil, English B.Sc. Criminology & Criminal Justice Science, B.B.A. Business Administration and PG Courses Computer Science, Commerce, Microbiology and Psychology and is affiliated to University of Madras.

GENESIS

The PERI PAPPATHI AMMAAL Educational Trust was founded in 2018. The vision of the trust is to provide a world class nodal centre of education where academics, communication, comprehension, visualization, practical application and leadership skills are imparted through encouraging research, training and development in technical and non-technical areas.

PERI College of Arts and Science is the brainchild of Mr. Saravanan Periasamy, the founding President & CEO of PERI Software Solutions Inc., a leading multinational IT Consulting and Services Company based in USA and the Chairman of the PAPPATHI AMMAAL Educational Trust. He is the source of inspiration and the pathfinder for this institution.

ABOUT THE DEPARTMENT

Over the year PG Department of Computer Science and Application have become the diamond in the crown of PERI College of Arts and Science. Every year we continue to produce humorous Software Professionals and the department has been their route to success. The department continues with the same objective of producing well-disciplined and talented Undergraduate and Postgraduate Software Professionals. The Department offers a B.Sc Computer Science as a three-year degree program and an M.Sc in Computer Science. The courses are designed in such a way that the students are given expertise in computer science, which is the most demanded technology of modern times. It is the most chosen course due to its course content with appropriate papers in computer science. The Department also offers Bachelor of Computer Applications (B.C.A), B.Sc. CS With Artificial Intelligence and B.Sc. CS With Data Science as a three years degree programme. The course content of the degree programme is fully on the application side of computer science which has the potential of making the learner as versatile in both i.e., subject and its applications. The job opportunities are much more and the student can pursue higher studies in computer science area. The Department has a well-established computer lab with new computers and the latest software. The library has journals, modern textbooks, and reference books.

Message from

HON'BLE CHAIRMAN



I am extremely happy to know that the PG Department of Computer Science and Computer Applications, PERI College of Arts and Science College, Chennai-48, India, is conducting a one-day International Conference on Cloud Computing and Data Science- ICCCDs'25, scheduled on 13th October 2025.

The conference record shows a total of five keynote address sessions by Eminent Academicians from Universities, Industry experts, and Researchers, and more than 400 delegates from all over the world participating in the conference. About 70 carefully reviewed research papers are going to be presented as oral presentations. I am confident that the conference deliberations will open new frontiers and enable collaboration between Academic and Research Institutions.

I want to express my deep appreciation to the Academic Director, the Principal and the Convener for their sincere efforts in bringing ICCCDs'25 a grand success.

I appreciate the organizing secretaries and members of the organizing committee for their excellent hard work to make the conference, and it is being considered as another feather added to the cap of PERI College of Arts and Science.

I am happy to extend my warmest felicitations to all associated with 'ICCCDS'25'.

I wish the Conference all Success.

Mr. SARAVANAN PERIASAMY

Hon'ble Chairman, PERI

Founder & CEO, PERI Software Inc., U.S.A.

Message from

COO



It gives me immense pleasure to welcome all the delegates to the International Conference Cloud Computing and Data Science- ICCCDs'25, scheduled on 13th October 2025 at PERI College of Arts and Science College, Chennai-600 048, organized by the PG Department of Computer Science and Computer Applications.

My special gratitude and appreciation are due to the various eminent personalities as they are primarily responsible for the content of the conference. I am glad to invite all the Academicians, Research Scholars, Faculties, Professional scientists and Industry experts to be a part of this fabulous event and express their conceptual ideas in our conference.

My sincere thanks to the conference organizing team members, who took so many efforts to make this event a grand success.

I hope that the conference will find valuable benefits to the society of students, researchers and our faculty members of Computer Science discipline.

Best Wishes.

Mr. SASIKUMAR VEERARAJAN
Chief Operating Officer, PERI

Message from

DIRECTOR- ACADEMICS



Greetings and hearty welcome to all.

I am immensely happy that PG Department of Computer Science and Computer Applications of our college is organizing “International Conference Cloud Computing and Data Science- ICCCDs’25”, scheduled on 13th October 2025.

Certainly, this much deserved conference brings together the notable scientists, research faculties and students from all over the globe to present and discuss about Research and Innovations in the field of Data Science and Cloud Computing.

About 400+ delegates are participating in the conference and more than 70 research papers are going to be presented.

My appreciation and congratulations to the Convener and the whole organising committee for their effort in conducting the Conference in a grand manner.

On behalf of PERI College of Arts and Science welcome you all and wish each one of you the very best.

I wish the conference a grand success.

Dr. A. GUNASEKARAN
Director, Academics

Message from

PRINCIPAL



It is a great honor for me to welcome you all to the “International Conference Cloud Computing and Data Science- ICCCDs’25” scheduled on 13th October 2025 in PERI College of Arts and Science, Chennai, India.

ICCCDS-25 is an accumulation to the efforts made by the Institute for bringing in concert Students, research scholars, illustrious academicians, and IT professionals with their precious deliverables on recent advancements in Cloud Computing and Data Science, on a common podium.

I am confident that this conference will provide an affable environment for both the academicians and researchers to exchange their views and ideas with the experts in the field.

I extend my warm greetings and congratulations to the Convener, conference organizing committee and participants, as well as my best wishes for all the presenters at the conference event.

On behalf of PERI College of Arts and Science welcome you all and wish each one of you the very best.

I wish the conference a grand success.

Dr. A. PURUSHOTHAMAN
Principal

Message from

CONVENOR



It gives me immense pleasure to extend a warm welcome to all the distinguished delegates, researchers, academicians, and industry experts gathered here for the International Conference on Cloud Computing and Data Science, organized by the PG Department of Computer Science and Applications, PERI College of Arts and Science.

In this era of rapid innovation, cloud computing and data science stand as pillars shaping the future of technology. This conference serves as a vibrant platform for sharing ideas, insights, and discoveries that drive progress in these dynamic fields.

I take great pride in serving as the Convener of this event and sincerely thank our respected Hon'ble Chairman, Mr. Saravanan Periyasamy, our Esteemed COO, Mr. Sasi Veerarajan, Dr. A. Gunasekaran, Director – Academics; and Respected Principal Dr. A. Purushothaman, for their constant support and encouragement.

Wishing everyone an inspiring and fruitful conference experience.

Mr. P. DINESH

HOD, PG Department Computer Science

Message from

CONVENOR



It gives me great pleasure to warmly welcome all participants to the International Conference on Cloud Computing and Data Science ICCCDs 2025.

This conference serves as a platform for researchers, academicians, and industry experts to exchange ideas, share innovations, and explore the latest trends in computational intelligence, data science, and emerging technologies.

In this digital era, data-driven approaches and computational methods are transforming the way we solve real-world challenges. Through engaging sessions and insightful keynote addresses, this conference aims to inspire collaboration and promote impactful research.

I take great pride in serving as the Convener of this event and sincerely thank our respected Hon'ble Chairman, Mr. Saravanan Periyasamy, our Esteemed COO, Mr. Sasi Veerarajan, Dr. A. Gunasekaran, Director – Academics; and Respected Principal Dr. A. Purushothaman, for their constant support and encouragement.

Wishing everyone a productive and inspiring conference experience.

Mrs. P. USHA

HOD, Department Computer Applications

About the Conference

ICCCDS'25 aims to offer an excellent opportunity to bring together professors, researchers, and scholars around the globe, a great platform to deliver the latest innovative research results and the most recent developments, and trends in cloud computing and data science. Invited speakers from around the globe will present at the meeting. The goal of ICCCDs'25 is to share visions and ideas, present research findings and get the latest information on emerging technologies

Call For Papers

Authors are invited to present their original and unpublished articles for oral poster presentations, on the below areas but are not limited to:

- Real-time Data Analytics on Streaming Data
- Multi-cloud and Hybrid Cloud Architectures
- Disaster Recovery and Fault Tolerance in Cloud Systems
- Blockchain for Secure Cloud Storage
- Generative AI Models for Data Augmentation
- Real-time Data Analytics on Streaming Data
- Predictive Analytics in IoT Sensor Networks
- Data Security and Privacy in Cloud-based ML Systems
- Machine Learning for Fraud Detection in Finance
- IoT Data Processing in Cloud for Smart Cities

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The Tamil Nadu Dr Ambedkar Law
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Dr. R. JAYSHREE
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Faculty of Engineering and Science
Department of Computer Science
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KEYNOTE / INVITED TALKS

PERI ICCDS' 2025





Generative AI in Research: Trends, Tools, and Ethical Implications

Dr. Josan D. Tamayo

Department Head, Information Technology
ICT Coordinator, CEU Malolos



Abstract:

Generative AI is rapidly advancing, with multimodal models like GPT-5, Claude 3.5, and Gemini 1.5 integrating text, image, and code reasoning. AI agents and embedded assistants streamline research workflows, supporting literature review, data analysis, and writing. Despite these benefits, challenges such as bias, copyright, data privacy, and AI-generated misinformation require careful ethical oversight. Researchers must disclose AI use, verify outputs, and maintain human judgment to ensure integrity. When used responsibly, generative AI empowers researchers, enhancing creativity, efficiency, and knowledge generation.

Keywords: Generative AI, Chat GPT





"A Comparative Study of Spectral Clustering and K-Means for Customer Segmentation in E-commerce"



Dr. E. Manigandan

Associate Professor, SOB-IT, Galgotias University,
Greater Noida,

Abstract:

Customer segmentation is a critical task in e-commerce to understand buyer behavior and improve marketing strategies. Clustering algorithms are widely used for this purpose, with K-Means being one of the most popular approaches due to its simplicity and speed. However, K-Means struggles with complex, non-linearly separable data. Spectral clustering, based on graph theory and eigenvalues of similarity matrices, has emerged as a powerful alternative capable of identifying clusters of arbitrary shape. This paper presents a comparative study of Spectral Clustering and K-Means in the context of customer segmentation using an open-source e-commerce dataset. The performance of both algorithms is evaluated using cluster validity metrics such as Silhouette Score and Davies-Bouldin Index. Our findings show that while K-Means performs well for spherical clusters, Spectral Clustering provides better separation and accuracy in more complex data distributions. This research highlights the importance of selecting the right clustering technique based on data characteristics to enhance customer insights in e-commerce platforms.

Keywords: Clustering, K-Means, E-Commerce



"Crop Insights from Above: Advancements in Indian Space-Based Remote Sensing for Agriculture"

Dr. Abhishek Pandey^{1*}, Dr. V. Ramesh²

^{1*}Assistant Professor, Manipal University Jaipur,

^{2*}Associate Professor, SCSVMV University, Kanchipuram
(Tamil Nadu),



Abstract:

With newly developed space-based platforms offering spatiotemporal insights into precision-level attributes like crop water consumption, fruits and vegetables health and condition, and crop response to management practices, satellite remote sensing holds great promise for bringing about a data-driven agricultural revolution. India is one of the leading space-faring countries in the world, leading the way in the use of satellite technology for several national and international purposes in addition to building satellites and launching those utilizing locally manufactured launch vehicles. India's space program has made significant strides in leveraging satellite technology for agricultural monitoring, contributing to the country's food security and rural development goals. Recently, sensors that use cutting-edge methods like Artificial Intelligence (AI) have become more important in the fields of remote sensing and intelligent agriculture. When it comes to assessing soil parameters, Remote Sensing (RS) technologies have many benefits over other approaches. These advantages include large-scale coverage, non-destructiveness, temporal surveillance, multidimensional abilities, and quick data collecting. In this work, we use the Vembanad-Kol-Wetland system in southwest India as a case study to demonstrate the use of remote sensing data to map flooded regions. A once-in-a-century flood produced by an exceptionally strong season of monsoons in August 2018 resulted in more than one million people being displaced and about 500 casualties in this area. We examine the use of current methods to map regions affected by flooding in the Lake Vembanad area by using the green, red, and near-infrared spectral reflectances from the MSI sensor aboard Sentinel-2. We demonstrate that flood maps can be generated from multi-spectral viewable imagery collected by remote sensing using the Modified Normalized Differences Water indexes and the Automatic Water Extraction Index, in addition to commonly used SAR-based techniques, to improve geographical coverage (from 12 to 5 days), even though the MSI sensor lacks cloud-penetrating capabilities.

Keywords: Remote Sensing, Artificial Intelligence (AI), Cloud-Penetrating, Water Index, Map, Sentinel-2, Techniques, Water Extraction, Lake Vembanad, Agriculture.



“Dare to Dream: Unlocking Potential Beyond Marks”

Dr. JAIMON ANTONY

Head of HR for India and Bangladesh at ESAB India Ltd,
Member of the Board of Directors at EWAC Alloys Limited.



Keynote Talk:

In a world driven by algorithms, rankings, and performance metrics, it's easy for students from modest academic backgrounds to feel left behind. This keynote by Dr. Jaimon Antony is a powerful reminder that potential is not measured by marks, but by mindset. Drawing inspiration from real-life stories — including the journey behind Soorarai Pottru — and his own experience in global HR leadership, Dr. Jaimon will challenge students to dream bigger, rise above limitations, and embrace their unique path in the tech-driven future.

Whether you're from science or arts, this session will ignite your belief that you belong in the world of innovation — and you have what it takes to shape it.



"Regulatory Challenges of AI in Cybersecurity: Balancing Autonomy, Ethics, and Accountability"

Dr. R. Deepalakshmi,

HOD, Department of CA

Director, Data Processing Centre School of Excellence in law

The Tamil Nadu Dr Ambedkar Law University, Chennai.



Abstract:

This chapter examines the legal and regulatory issues raised by AI's application to cybersecurity, with a particular emphasis on AI-driven cyberattack and defensive approaches. It looks at the existing legal frameworks, identifies any weaknesses, and makes recommendations for how artificial intelligence (AI) could be handled in the cybercrime space. Artificial intelligence (AI) is bringing both potential and legal challenges to cybersecurity as it continues to transform it, especially in terms of AI-driven cyber offensive and defensive systems. The legal ramifications of AI's application in cybersecurity are examined in this article, combined with the appropriateness of current frameworks, ethical challenges with AI-operated cyberattacks, and accountability concerns with AI's decision-making autonomy. This research emphasizes the need for updated legal frameworks to address the rapid developments in AI-driven cybersecurity by examining present laws and discovering significant gaps.

According to the CERT Guide to Coordinated Vulnerability Disclosure, it is "a set of conditions or behaviors that allows the violation of an explicit or implicit security policy." This report uses this definition. Software flaws, configuration or choice of design, unplanned system interactions, and environmental changes can all result in vulnerabilities.

These approaches are vulnerable to a wide range of manipulations that might lead to errors, reveal model parameters, impair performance, or infer personal data from training datasets. Those AI models, including those that have been used in public circumstances, have been shown to have significant weaknesses by researchers.

Keywords: AI, Cyberattacks, CERT



"Improved Parkinson disease prediction with hybrid Machine learning techniques"

Dr. Jayashree R

Department of Computer Science, Faculty of Engineering
and Science, Dili Institute of Technology, Dili, Timor-Leste.

ORCID: 0000-0002-0150-7095



Abstract:

This paper establishes a complete approach developed to predict Parkinson's Disease using a hybrid approach that combines Machine Learning (ML) and Deep Learning (DL) techniques. The model is designed to provide accurate and efficient predictions, supporting healthcare professionals and individuals in early diagnosis and timely medical intervention. The primary objective is to foster an intelligent and accessible tool capable of exploring patient data to uncover potential health risks. Beforehand detection is crucial for preventing complications and improving treatment outcomes. By suggesting prelude insights through a robust data-driven approach, the model bridges the variance between symptoms and specialist analysis. The architecture gives precedence to modularity, performance, and scalability. The models trained on relevant medical datasets to produce high-diligence predictions. Through advanced data-driven techniques, this paper augments the reliability of initial illness recognition tools and provides artificial intelligence (AI) assisted diagnostic validation to the medical community. The amalgamation of Logistic Regression (ML) and Deep Neural Networks (DL) enables the scheme to uncover complex samples in medical data, substantially improving prediction precision with a 98% F1-score. Finally, the paper targets to enhance healthcare technology by harnessing AI to develop a robust decision-support tool that demotes diagnostic delays and promotes early intervention.

Keywords: Supervised Feature Selection, Deep Neural Networks, Logistic Regression, Parkinson's Disease Prediction System



RESEARCH PAPERS AND PRESENTATIONS

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A COMPARATIVE STUDY OF MACHINE LEARNING APPROACHES FOR FRAUD DETECTION IN FINANCIAL SERVICES

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Mr.M.KarthickMCA.,Assistant Professor Department of Computer Science,
Sri Balaji Arts and Science college, Kolapakkam

ABSTRACT

Financial fraud has become more and more sophisticated, costing financial institutions billion annually. Traditional rule-based methods often fail to keep pace with rapidly evolving fraud patterns. Machine learning (ML) techniques provide adaptive, data-driven solutions capable of detecting complex anomalies in financial transactions. The process of identifying suspicious activities like credit card theft to investment scams is defined as Fraud detection. In addition to financial losses, fraudulent activities can cause reputational damage, business interruptions and lost productivity. Insurance providers, financial institutions and others can face regulatory mandates to detect and prevent fraud. Challenge factors like evolving Fraud Tactics, Data Overload, False Positives and Negatives, Integration Issues exists in fraud detection. While traditional rule-based systems lack the adaptability required to detect novel fraud strategies. Machine learning has emerged as a promising approach to address these gaps by learning complex relationships from transactional data. This paper presents a comparative study of different ML approaches for fraud detection, including supervised models, unsupervised anomaly detectors, and deep learning techniques. Using benchmark datasets Insurance Claim Fraud Dataset, we evaluate models based on detection accuracy, precision-recall trade-offs, computational efficiency, and interpretability. Results highlight the accuracy of each method Logistic Regression (78%), Random Forest (86%), Gradient Boosting (89%) and Neural Networks (91%) providing insights for both researchers and practitioners in designing effective fraud detection systems.

Keywords: Fraud detection, financial fraud, anomaly detection, supervised learning, deep learning, predictive analytics, generative AI, financial security.

A PRIORITY-OPTIMIZED AGGREGATION FRAMEWORK FOR MULTI-EXPERT MULTI-CRITERIA NEUTROSOPHIC ASSIGNMENT PROBLEMS

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ABSTRACT

This chapter introduces a Priority-Optimized Aggregation framework for addressing Multi-Expert Multi-Criteria Neutrosophic Assignment Problems under uncertain decision-making environments. The proposed framework integrates the opinions of multiple experts and diverse evaluation criteria through the Priority-Optimized Mean operator, enabling balanced treatment of truth, indeterminacy, and falsity in neutrosophic assessments. A weighted score function is employed to transform aggregated neutrosophic values into comparable decision scores, ensuring flexibility in prioritizing different dimensions of uncertainty. The mathematical formulation is generalized to handle any number of experts, criteria, and neutrosophic inputs. To demonstrate the practical utility, a construction-related task allocation problem is presented with complete data, detailed computations, and optimization of assignments. The results highlight the model's effectiveness in producing rational, transparent, and uncertainty-aware assignments, making it suitable for engineering, management, and real-world decision-making applications.

KEYWORDS: Neutrosophic Assignment Problem, Multi-Expert, Multi-Criteria, Priority-Optimized Mean, Uncertainty, Aggregation, Decision-Making

A STRUCTURAL ALGORITHMIC MODEL FOR PRICE MOVEMENTS

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ABSTRACT

In today's financial markets, computerized systems have made placing orders and accessing real-time data easier than ever. This has given rise to a variety of trading approaches, from price action and support/resistance to chart patterns and auction-based theories. Yet, despite these tools, most retail participants continue to struggle. SEBI's FY25 survey found that nearly 91% of traders recorded losses, amounting to about 1.6 trillion INR. The main reasons are emotional factors, poor use of data, lack of clear entry and exit rules, and overexposure. Simply having market data isn't enough—it's about knowing how to use it effectively. That's where algorithmic systems come in. By following predefined rules, removing emotions from decision-making, and enabling strategy back-testing before going live, they help traders stay disciplined and consistent. As markets grow more complex, such systems are becoming essential for sustainable success.

Key words: Computerized systems, Emotional factors, FY25, Algorithmic systems, Chart patterns, Strategy back-testing.

A STUDY ON THE EVOLUTION AND APPLICATIONS OF GENERATIVE AI TECHNOLOGIES

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ABSTRACT

Artificial intelligence (AI) comprises various technologies that enable machines to perform tasks commonly linked to human intelligence, including sensing, interpreting, making decisions, and learning. A specialized branch of AI, generative AI, focuses on creating novel content—such as text, images, sounds, and synthetic data—instead of just processing or classifying pre-existing information. Generative AI is an innovative type of artificial intelligence that rapidly produces content based on given inputs. It can acquire knowledge in diverse areas including human languages, programming, art, chemistry, biology, and other complex disciplines. By leveraging its learned information, generative AI can tackle new challenges. Organizations can apply generative AI across multiple domains, including developing chat bots, creating media, designing products, and driving innovation. Educators and professionals from a various discipline can greatly benefit from developing a solid understanding of the basic principles of artificial intelligence (AI) and generative artificial intelligence (Gen-AI). This paper aims to introduce these fundamental concepts by first covering essential topics such as artificial intelligence, machine learning, deep learning, artificial neural networks, and large language models (LLMs), which provide the foundation for understanding generative AI. The discussion then extends to the practical uses and potential effects of generative AI in both business and educational settings. Lastly, the paper highlights some of the current challenges and issues associated with generative AI technologies.

Keywords

Generative AI, Artificial Intelligence, Machine Learning, Deep Learning, Generative Models, AI Applications, Large Language Models (LLMs)

A SURVEY ON CLOUD SECURITY ISSUES AND TECHNIQUES

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ABSTRACT

Currently, cloud computing signifies an innovative paradigm in the field of computer science. Cloud computing encompasses an assembly of resources and services provided via the network or internet. Cloud computing encompasses different computational methodologies, including grid computing and distributed computing. Currently, cloud computing is utilized in both industrial and academic sectors. The cloud provides users with virtual resources through the internet. As the domain of cloud computing enlarges, innovative techniques are emergent. The extension of cloud computing environments also intensifies security experiments for cloud developers. Cloud users stock their data on the cloud; therefore, inadequate security can destabilize user trust. This article will scrutinize several cloud security challenges, including multi-tenancy, elasticity, and availability. The article also inspects current security techniques and methodologies for a secure cloud. This document will apprise researchers and professionals about numerous security dangers, models, and proposed methods.

Keywords—Cloud Computing, Cloud Security, Security Threats, Security Techniques, Cloud Security Standards.

A DUAL-STREAM GRAPH CONVOLUTIONAL AND TRANSFORMER ARCHITECTURE FOR ACCURATE SKELETON-BASED HUMAN ACTION RECOGNITION

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ABSTRACT

Achieving high-accuracy skeleton-based human action recognition has long relied on sequential frameworks that combine Graph Convolutional Networks (GCNs) with attention mechanisms. However, these methods often treat the human skeleton as a single, rigid entity, overlooking the importance of correlated but indirectly connected body parts—thereby reducing recognition precision. To address this issue, we propose a novel parallel dual-stream network architecture, termed SA TDGFormer, which integrates GCN and Transformer models to jointly extract both local and global spatio-temporal features. The model consists of two complementary streams: a spatiotemporal GCN stream that captures the complex topological structure and motion patterns of the human skeleton, and a spatiotemporal Transformer stream that models long-range inter-joint dependencies. A late fusion strategy is employed to combine the outputs from both streams, allowing each to enhance the other's representations and improve overall recognition capability. Extensive experiments conducted on three benchmark datasets—NTU RGB+D 60, NTU RGB+D 120, and Kinetics-Skeleton—demonstrate that the proposed method achieves superior performance, surpassing existing frameworks by 1–5% on the NTU RGB+D 60 dataset. These results validate the effectiveness of the proposed architecture in capturing comprehensive motion dynamics and improving the accuracy of skeleton-based action recognition.

Keywords : Action recognition, Graph convolutional networks, Transformer

AI-DRIVEN AUTONOMOUS RESILIENCE AND EDGE-DISPERSAL FOR CLOUD DISASTER RECOVERY

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ABSTRACT

Traditional disaster recovery in cloud systems depends on static geo-redundancy and human-driven mechanisms that activate only after a disaster occurs. To address this limitation, this paper proposes an **AI-driven autonomous resilience framework** that integrates distributed data centers, real-time environmental monitoring, and predictive maintenance. The framework introduces **Intelligent Load-Shifting and Data Migration (ILSDM)**, where AI dynamically analyzes disaster indicators such as seismic activity, severe weather, and wildfire data to proactively migrate critical workloads and datasets to resilient nodes, minimizing disruption and reducing data loss. Additionally, **Predictive Infrastructure Maintenance (PIM)** leverages IoT sensors and AI analytics to forecast failures in essential backup components such as cooling systems, power generators, and batteries, enabling precautionary actions before emergencies occur. By transforming disaster recovery from a reactive process to an intelligent and self-managing system, this approach enhances **fault tolerance**, ensures **service continuity**, and strengthens the overall resilience of cloud infrastructure during natural calamities.

Keywords: Static Geo-Redundancy, AI-Driven Autonomous Resilience Framework, Real-Time Environmental Monitoring, Intelligent Load-Shifting and Data Migration (ILSDM), Predictive Infrastructure Maintenance (PIM).

AI-DRIVEN WEATHER-BASED CROP PREDICTION SYSTEM WITH SMS ALERTS FOR SUSTAINABLE AGRICULTURE

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ABSTRACT

Agriculture is a crucial sector for ensuring food security and sustaining livelihoods, yet it remains highly vulnerable to changing weather conditions. Uncertainty in rainfall, temperature fluctuations, and delayed decision-making often result in poor crop selection and reduced productivity. To address this issue, the integration of Machine Learning (ML) and Artificial Intelligence (AI) offers effective solutions for predicting crop suitability based on weather patterns. This project presents a weather-based crop prediction system utilizing a Kaggle dataset of over 300,000 records containing parameters such as rainfall, temperature, and humidity. By applying machine learning algorithms, the system identifies the most suitable crops for a given season and location, thereby enabling farmers to make informed decisions. A significant feature of the proposed system is the SMS alert service, which delivers predictions directly to farmers' mobile phones, ensuring accessibility even in rural areas with limited internet connectivity. The proposed approach not only supports precision farming but also enhances resilience against climate risks and promotes sustainable agriculture. By combining predictive analytics with mobile-based communication, the system minimizes crop losses, improves yield, and empowers farmers with actionable insights. This work highlights the potential of AI-driven solutions in transforming traditional agriculture into a data-driven and farmer-friendly practice.

Keywords: Agriculture prediction, Machine Learning, Weather forecasting, Crop recommendation, SMS alert system, Precision farming, Sustainable agriculture.

AI-POWERED IOT CCTV FOR REAL-TIME ACCIDENT DETECTION IN SMART CITIES

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ABSTRACT

Smart cities aim to enhance public safety and optimize emergency response systems using advanced technologies. This research proposes an AI-powered IoT CCTV framework for real-time accident detection and emergency management. The system continuously monitors city-wide traffic cameras, leveraging computer vision and deep learning algorithms to detect accidents and classify injury severity. Upon detecting an incident, the system automatically identifies the nearest suitable hospital with the required medical facilities and sends instant notifications via calls, messages, and emails to ambulance services and authorities. Additionally, the framework incorporates legal documentation automation, generating evidence-ready reports for police and legal compliance. By integrating real-time AI analytics, IoT infrastructure, and cloud-based processing, this approach aims to significantly reduce emergency response times, improve accident outcomes, and streamline public safety operations in smart cities.

Keywords: Smart Cities, IoT, AI, CCTV, Accident Detection, Emergency Response, Computer Vision, Deep Learning, Cloud Computing, Legal Automation

AN AGENTIC AI FRAMEWORK FOR REAL-TIME OPTIMIZATION OF RAPID TRANSIT IN DEVELOPING CITIES

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ABSTRACT

Transportation plays a major role in developing cities like Dar es Salaam, Tanzania. The number of vehicles traveling in Dar es Salaam continues to increase daily, largely due to its role as a major port serving several landlocked countries. Consequently, the Dar es Salaam Rapid Transit (DART) system faces challenges of operational inefficiency, safety concerns, and revenue leakage. Although the Tanzanian government has taken measures such as constructing roads, improving traffic control, and implementing digitalized billing systems, further solutions are needed. This paper proposes an Agentic AI framework that integrates autonomous software agents across fuel and maintenance management, GPS fleet tracking, passenger monitoring, billing systems, and traffic operations. The architecture combines on-bus edge intelligence with cloud-based coordination to enable real-time perception, decision-making, and learning. Privacy-preserving computer vision, predictive maintenance, anomaly detection, and revenue reconciliation are supported by these specialized agents.

Keywords

Agentic AI, Intelligent Transportation Systems, Dar es Salaam, Predictive Maintenance, Public Transport, Smart Mobility, Edge-to-Cloud Integration

A MACHINE LEARNING FRAMEWORK FOR PREDICTING CHILDHOOD MALNUTRITION

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ABSTRACT

Malnutrition continues to be a critical public health issue, especially in developing countries where children under the age of five are most at risk. Early identification and intervention are vital to preventing morbidity, mortality, and long-term developmental challenges associated with under nutrition. With the growing advancement of data-driven technologies, Machine Learning (ML) presents powerful tools for predicting malnutrition by analyzing intricate patterns across health, demographic, and socio-economic factors. This study presents an optimized machine learning-based framework for predicting child malnutrition. The model utilizes comprehensive datasets encompassing anthropometric indicators, dietary patterns, household income, sanitation conditions, and healthcare accessibility. The machine learning algorithms used for Childhood Malnutrition prediction are Logistic Regression, Random Forest, Support Vector Machine, Naïve Bayes, and Gradient Boosting—were trained and assessed for performance. The prediction of childhood malnutrition using the Logistic Regression algorithm resulted in an accuracy of 82% .By facilitating early detection the proposed predictive framework supports targeted nutritional interventions, efficient resource distribution, and progress toward achieving Sustainable Development Goal 2: Zero Hunger.

Keywords— Machine Learning, Malnutrition, Prediction, Public Health

AN EMPIRICAL APPROACH OF COTTON LEAF DISEASE DETECTION USING CNN

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ABSTRACT

Precision agriculture aims to improve agricultural productivity by combining technology and farming. The productivity of the cotton is mostly affected by leaf diseases, if we predict these diseases at early stage, it helps the farmers to improve productivity. We put forward a novel method for cotton leaf disease detection that works on hybrid dataset which comprises of images from the kaggle dataset and real-time images. Deep learning models VGG16 and VGG19 are applied on this hybrid dataset for disease detection of cotton leaves. This research will not only contribute to improve crop health but also be a valuable resource for farmers. A systematic comparison of the VGG16 and VGG19 models reveals their functional differences in disease detection. VGG16 and VGG19 has achieved accuracy of 94% and 95% in disease detection

Keywords: Hybrid Dataset, Vgg16, Vgg19, CNN, HealthyLeaf, Diseased leaf

AI-DRIVEN CROP RECOMMENDATION SYSTEM USING MACHINE LEARNING

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ABSTRACT

Using machine learning (ML) techniques, the AI-Driven Crop Recommendation System makes data-driven crop recommendations based on soil, climate, and environmental parameters. These systems are essential for maximizing agricultural yields, guaranteeing food security, and encouraging sustainable farming methods as global agriculture deals with issues including resource scarcity, climate change, and shifting consumer demands. This system analyses and forecasts the best crops for particular areas by combining data from multiple sources, such as historical weather patterns, soil composition, moisture content, and geographic data. The technology efficiently processes complicated facts and offers precise crop suggestions by leveraging cutting-edge machine learning methods including decision trees, random forests, and neural networks. These suggestions are flexible and responsive to seasonal fluctuations, unanticipated weather events, and real-time environmental changes, assisting farmers in making prompt, well-informed decisions. The system promotes optimal farming practices by taking into account factors like fertilizer usage, pest management, and irrigation requirements in addition to crop selection. It has an easy-to-use interface that lets farmers get personalized advise on their devices and is available on both online and mobile platforms. This AI-driven solution seeks to boost agricultural output, lower crop failure risks, and support sustainable agricultural development across a range of climatic zones by providing farmers with fast, accurate information.

Keywords: crop recommendation, early detection, growth stages, threats, sustainability, computer vision, machine learning, remote sensing, predictive analytics, resource optimization, and yield potential.

AN OPTIMAL INTELLIGENCE-BASED MODEL BUILDING FOR DETECTING INFECTIOUS DISEASES

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ABSTRACT

Infectious diseases are caused by microorganisms belonging to the class of bacteria, viruses, fungi, or parasites. These pathogens are transmitted, directly or indirectly, and can lead to epidemics or even pandemics. Artificial Intelligence (AI) combined algorithms including Machine Learning and Deep Learning have emerged as a potential solution to analyse diverse datasets and face challenges posed by infectious diseases. Computer-aided detection (CAD) using convolutional neural networks (CNN) has gained prominence for diagnosing tuberculosis (TB) and other infectious diseases such as COVID-19, HIV, and viral pneumonia. Artificial neural networks (ANN), recurrent neural networks (RNN), support vector machines (SVM), multilayer neural networks (MLNN), CNN, long short-term memory (LSTM), and random forests (RF) are among the models discussed. The review emphasizes the potential of AI to enhance the accuracy and efficiency of diagnosis, treatment, and Detection of infectious diseases, highlighting the need for further research and development in this area. The transformative potential of AI in infectious diseases is emphasized, addressing gaps in rapid and accurate disease diagnosis, surveillance, outbreak detection and management, and treatment optimization. The goal is achieved by developing a mathematical model that identifies potential infectious diseases from medical images, adopting a multi-criteria decision-making approach.

Keywords: convolutional neural networks, medical images, artificial intelligence, diagnosis.

APPLICATION OF CLOUD COMPUTING IN HEALTHCARE – A COMPREHENSIVE STUDY

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ABSTRACT

Cloud computing has emerged as a transformative force in healthcare and biomedical sciences, offering scalable, on-demand resources for managing vast amounts of data. This review explores the integration of cloud computing within these fields, highlighting its pivotal role in enhancing data management, security, and accessibility. We examine the application of cloud computing in various healthcare domains, including electronic medical records, telemedicine, and personalized patient care, as well as its impact on bioinformatics research, particularly in genomics, proteomics, and metabolomics. To propose a categorization methodology with five different categories — Cloud-based framework development in healthcare, virtual machine placement in healthcare, cloud-based green healthcare, security in cloud-based healthcare, and broker development in healthcare. This paper provides a review of some proposed cloud-based e-healthcare architectures edge along with issues in both technologies and the crucial reasons of enhancing forward to a cloud-based e-healthcare.

Keywords : Healthcare, bio medical, data management, telemedicine, bio informatics, virtual machine.

ARTIFICIAL INTELLIGENCE IN EDUCATION FOR STUDENT PLACEMENTS

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ABSTRACT

Students' Performance Evaluation for Placements plays a vital role for educators to know the weaknesses and strengths of a student for his growth. By receiving the academic scores such as his CGPA, internship score, extracurricular activities score, and his backlogs, attendance percentage, soft skills and certifications, helps us to analyze the overall score for his placement. By utilizing data analytics and machine learning algorithms, we can track and assess the participation and growth of students. This paper provides an overview for both students and educators about the academic and non-academic achievements. It uses algorithms to predict students' academic performances at various stages and identifies the improvements needed for their placements.

Keywords: Artificial Intelligence, Machine Learning, Linear Regression, K-means.

ARTIFICIAL INTELLIGENCE IN INDIAN AGRICULTURE: TRENDS, CHALLENGES, AND FUTURE DIRECTIONS

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ABSTRACT

The next generation demonstrates significant interest in agriculture and anticipates increased productivity through the adoption of advanced technologies. Artificial intelligence (AI) is particularly prominent, as it fosters an ecosystem that benefits all stakeholders. Technological advancements should establish a comprehensive framework that supports farmers throughout the cultivation process, from land preparation to seed selection and fertilization. These tools enable the identification of optimal crops for each season. Additionally, technology facilitates market analysis and demand forecasting, allowing farmers to achieve higher yields. AI-driven agriculture enhances productivity and profitability for individual farmers. Direct-to-consumer markets, enabled by technology, improve product quality and help stabilize prices. The integration of technology in agriculture is likely to attract the next generation to agricultural careers. Predictive analytics further assist farmers in understanding market demand. Ultimately, AI and related fields are expected to play a central role in the development of future agricultural systems

Keywords: Artificial Intelligence, Demand Forecasting, Predictive analytics

A SCALABLE EXPLAINABLE AI FRAMEWORK USING CLUSTERING ENSEMBLE WITH INTERSECTING K-MEANS FOR BIG DATA INTERPRETATION

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ABSTRACT

Explainable Artificial Intelligence (XAI) has emerged as a crucial field in ensuring the transparency and trustworthiness of machine learning models. However, scalability and interpretability remain significant challenges in handling massive, heterogeneous datasets. This study introduces a novel distributed framework named **Clustering Ensemble with Intersecting K-Means (CEIK)** to enhance interpretability in large-scale data environments. The proposed CEIK model integrates both local and global reasoning to provide comprehensive insights into model predictions while maintaining computational efficiency. By leveraging distributed big data technologies and ensemble learning, CEIK achieves sub-linear scalability and high-quality explanations across diverse datasets. Experimental results demonstrate that the model improves explanation accuracy, clustering purity, and interpretability, thereby supporting trustworthy and scalable AI systems applicable across finance, healthcare, and cybersecurity domains.

Keywords: Explainable Artificial Intelligence, Big Data Analytics, Clustering Ensemble, Machine Learning, Interpretability

A STRUCTURAL ALGORITHMIC MODEL FOR PRICE MOVEMENTS

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ABSTRACT

In today's financial markets, computerized systems have made placing orders and accessing real-time data easier than ever. This has given rise to a variety of trading approaches, from price action and support/resistance to chart patterns and auction-based theories. Yet, despite these tools, most retail participants continue to struggle. SEBI's FY25 survey found that nearly 91% of traders recorded losses, amounting to about 1.6 trillion INR. The main reasons are emotional factors, poor use of data, lack of clear entry and exit rules, and overexposure. Simply having market data isn't enough—it's about knowing how to use it effectively. That's where algorithmic systems come in. By following predefined rules, removing emotions from decision-making, and enabling strategy back-testing before going live, they help traders stay disciplined and consistent. As markets grow more complex, such systems are becoming essential for sustainable success.

Key words: Computerized systems, Emotional factors, FY25, Algorithmic systems, Chart patterns, Strategy back-testing.

BLOCKCHAIN-BASED SECURE DATA SHARING FOR CLOUD APPLICATIONS

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ABSTRACT

The rapid growth of cloud computing has enabled scalable storage and global access to digital resources. However, challenges in data privacy, integrity, and trust remain major obstacles to secure data sharing in cloud environments. This paper presents a blockchain-based framework that integrates decentralized trust, smart contracts, and cryptographic techniques to enhance secure data sharing in cloud applications. The proposed model employs encryption for confidentiality, blockchain for immutability, and smart contracts for automated access control. This framework addresses issues of unauthorized access, tampering, and insider threats, while ensuring transparency and auditability.

Keywords— Blockchain, Cloud Computing, Secure Data Sharing, Smart Contracts, Cryptography.

BREATHESAFE: INTERNET OF THINGS -POWERED AIR QUALITY MONITORING AND PUBLIC ALERTS

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ABSTRACT

Air pollution is becoming a bigger problem for people living in cities, and it is harming both their health and the environment. There is a growing need for real-time air quality checks at the street level to help protect people's health and enable better decision-making in smart cities. Existing monitoring systems, such as government stations and platforms like AQICN, provide reliable but generalized city-wide data through fixed sensors. However, these systems do not offer detailed, localized information or specific warnings and predictions for certain areas. The BreatheSafe system addresses these challenges by using low-cost IoT sensors and a cloud platform called Thingier.io to collect continuous environmental data. Simultaneously, it gathers weather information from the OpenWeather API, which helps create a comprehensive dataset for analysis. A Random Forest Regressor model, developed using Python, is employed to predict the Air Quality Index (AQI) based on real-time parameters such as gas levels, temperature, humidity, and weather conditions. This system can send timely alerts when air quality becomes or is predicted to become harmful, enabling people to take precautionary measures to protect their health. Future improvements may include the integration of GPS-enabled mobile sensors for enhanced coverage, real-time AQI heatmap generation, and connectivity with government APIs for automatic emergency alerts. These upgrades would make BreatheSafe a robust tool for improving environmental health in urban areas.

Keywords: Air Quality Index (AQI), IoT Sensors, Random Forest, Thingier.io, OpenWeather API

CLOUD COMPUTING IN AGRICULTURE

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ABSTRACT

Use of Cloud computing technology in agricultural areas has greater chance in the overall development of India. An effective implementation of cloud computing is encouraging in agricultural sector. Cloud Computing is emerging today as a commercial infrastructure that eliminates the need for maintaining expensive computing hardware, software, Information technology, staff, infrastructure, recourses and their maintenance. Cloud computing is a network-based environment that focuses on sharing computations, Cloud computing networks access to a shared pool of configurable networks, servers, storage, service, applications & other important computing resources. Indian agriculture needs to be modernised through the use of technology to increase productivity, wealth distribution, and cost management. In this article, we propose a comprehensive smart agriculture paradigm based on the following technological advances: Sensors, Internet of Things (IoT), Cloud Computing, Mobile Computing, and Big Data Analytics.

CLOUD-ENABLED IOT DATA PROCESSING IN SMART CITIES: OPPORTUNITIES, CHALLENGES, AND FUTURE DIRECTIONS

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ABSTRACT

The development of smart cities is increasingly dependent on the integration of Internet of Things (IoT) technologies, which enable the collection of vast amounts of heterogeneous data from urban environments. Sensors, smart devices, and connected infrastructures generate continuous data streams related to traffic management, healthcare monitoring, energy distribution, waste management, and public safety. Processing this massive and dynamic data efficiently is essential to provide meaningful insights and support real-time decision-making. Cloud computing has emerged as a powerful platform for IoT data processing due to its elasticity, scalability, and ability to handle high-volume and high-velocity data. By offering distributed storage, advanced analytics, and machine learning capabilities, cloud platforms allow city administrators and service providers to optimize resources, predict urban challenges, and enhance the quality of life for citizens. Furthermore, cloud-based IoT data processing supports interoperability across multiple domains, enabling seamless integration of diverse applications and services within a smart city ecosystem. However, several challenges remain, including data privacy, latency reduction, network congestion, and the dependency on stable connectivity. To address these issues, hybrid models that combine cloud computing with edge or fog computing are being explored to bring processing closer to the data source while maintaining the advantages of centralized cloud resources. This paper highlights the significance of cloud-enabled IoT data processing in smart cities, analyses key opportunities and limitations, and discusses future research directions toward developing more sustainable, secure, and resilient urban infrastructures.

Keywords: Smart Cities, Internet of Things (IoT), Cloud Computing, Real-Time Data Processing, Big Data Analytics, Edge Computing, Interoperability.

COMPARATIVE PERFORMANCE ANALYSIS OF FUZZY NUMBER WITH VARIOUS RANKING IN TRANSPORTATION PROBLEM

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ABSTRACT

This study focuses on efficiently minimizing transportation cost in a real-world supply chain where the cost of moving goods from warehouses to showrooms is uncertain. The research employs a fuzzy methodology. The uncertain transportation costs are initially represented using various types of fuzzy numbers, including triangular, trapezoidal, and pentagonal. The core step of this approach is to convert these vague fuzzy costs into definite crisp numbers using specialized ranking functions. Once the costs are made concrete, the classical transportation problem can be solved using a sequential application of established optimization techniques. The process begins with Vogel's Approximation Method (VAM) to find the initial feasible solution. This solution is further optimized using the Modified Method and compared with the Row Column Reduction Method. The final output involves determining the attainable total cost and conducting a comparative analysis of the results obtained.

Keywords: Fuzzy Set, Triangular, Trapezoidal and Pentagonal Fuzzy Number, Maleki's Ranking, Yager's Ranking, Robust Ranking, Pascal Triangular Graded Mean Ranking

CONTEXTUAL AND EMOTION –AWARE CYBER BULLYING DETECTION IN CODE-MIXED SOCIAL MEDIA CONVERSATIONS USING NLP AND DEEP LEARNING

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ABSTRACT

In the current era, cyber bullying continues to be a major problem with severe consequences. It may lead to emotional turmoil, decreased self-worth, isolation from social interactions, poor academic results and mental health challenges, such as anxiety and depression. This research paper presents an innovative approach to detect cyber bullying on social media platforms by integrating Natural Language Processing (NLP) and Deep Learning techniques. This study tackles the growing issue of cyber bullying by utilizing contextual and emotion-aware models applied to code-mixed social media data, encompassing comments, tweets, and messages. The proposed detection model preprocesses text data using NLP methods to extract meaningful linguistic features and applies various DL techniques including CNN, LSTM and Hybrid approach to identify bullying behavior effectively. The performance of the proposed model is evaluated using key classification metrics, including accuracy, precision, recall, and F1 score. The experimental results demonstrate the efficiency and accuracy of this hybrid approach with some algorithms achieving detection accuracies above 90%. In addition to the binary classification of bullying content, this study explored multilingual and bilingual detection frameworks and emphasized early intervention strategies to prevent such harm. This study contributes to creating safer online environments by enhancing the real-time identification and mitigation of cyber bullying using computing intelligence.

Keywords: Cyber bullying, Deep Learning, Natural Language Processing (NLP), Social Media, CNN, LSTM etc.

CLOUD-FIRST MOBILE OPERATING SYSTEM WITH AI-DRIVEN RESOURCE MANAGEMENT

Authors

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ABSTRACT

A cloud-first mobile operating system removes traditional storage limitations by offloading all user data, applications, and media to cloud infrastructure. At startup, the system executes the **OS Boot Algorithm**, which initializes core hardware, authenticates the user, and mounts cloud storage as a virtual file system. Local storage is reserved strictly for essential services like phone calls and real-time camera capture, governed by the **Phone & Camera Handler Algorithm**. Captured media is temporarily stored and then uploaded via the **Media Upload Algorithm**, which uses network and battery-aware logic to sync files efficiently. Applications and documents are accessed on demand through the **App Launch** and **File Access Algorithms**, with a **Cache Management Algorithm** monitoring device storage and offloading rarely used data. Machine learning models are integrated into the **Smart Prediction Algorithm**, which forecasts the next likely apps or files a user will need and preloads them to improve responsiveness. A **Cloud Sync Engine Algorithm** maintains continuous synchronization between cloud and device, adjusting sync timing dynamically based on usage patterns and resource availability. Deep learning enhances media compression and voice interactions through the **Media Optimization Algorithm** and **Voice Interface Algorithm**, reducing bandwidth usage while improving user experience. An **Anomaly Detection Algorithm** continuously monitors system behavior to flag suspicious activity, securing cloud interactions. In offline scenarios, the **Offline Mode Algorithm** ensures that essential cached apps and services remain accessible. Collectively, these algorithms form a seamless and intelligent OS experience that enables ultra-low-cost smart phones with minimal local storage and near-unlimited, AI-optimized cloud capabilities.

Keywords: OS Boot and Initialization, File Access and Storage Management, Cloud Sync Engine, Security and Encryption.

COMBINING ENSEMBLE MACHINE LEARNING AND SHAP INTERPRETATIONS FOR EXPLAINABLE PCOS RISK STRATIFICATION

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ABSTRACT

Polycystic Ovary Syndrome (PCOS) is a common yet multi factorial endocrine disorder that affects women of reproductive age and is associated with irregular menstrual cycles, hormonal imbalance, infertility, and long-term metabolic complications. Early diagnosis and accurate risk assessment remain challenging due to the heterogeneous presentation of symptoms and the complex interactions of lifestyle, genetic, and clinical factors. To address this challenge, this paper proposes an explainable machine learning framework for PCOS risk stratification that integrates ensemble models with SHapley Additive exPlanations (SHAP). Secondary data sourced from Kaggle, encompassing demographic, lifestyle, and clinical attributes, is used to develop and evaluate predictive models. Three algorithms—Support Vector Machine (SVM), Random Forest, and XGBoost—are trained and compared for their ability to classify risk levels. To enhance both accuracy and interpretability, SHAP-based feature importance analysis is employed to identify the ten most influential predictors. The models are then re-evaluated on this reduced feature subset to examine robustness and generalizability. Beyond predictive performance, the incorporation of SHAP ensures that the framework provides transparent, instance-level explanations, allowing a deeper understanding of how specific features contribute to individual predictions. This interpretability bridges the gap between complex machine learning models and their clinical application, fostering trust among healthcare practitioners. The findings of this study underscore the potential of combining ensemble learning with explainable AI techniques in building reliable, interpretable, and clinically meaningful decision-support systems for PCOS risk stratification.

DATA SECURITY AND PRIVACY FRAMEWORK FOR CLOUD-BASED MACHINE LEARNING SYSTEMS

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ABSTRACT

Cloud-based machine learning (ML) systems have become the backbone of modern applications across healthcare, finance, and industry. However, their reliance on sensitive datasets and multi-tenant cloud infrastructures raises significant **data security and privacy concerns**. While cloud providers offer strong infrastructure protections, these are insufficient against **ML-specific attack vectors** such as **membership inference, model inversion, data poisoning, and model extraction**, which directly target models and training pipelines. Addressing these challenges requires a holistic approach that integrates both **algorithmic defenses** and **operational safeguards**. This paper introduces a **comprehensive security and privacy framework** for cloud-based ML systems. At the algorithmic level, we survey state-of-the-art defenses—**differential privacy, federated learning, secure multiparty computation, and adversarially robust training**—and evaluate their trade-offs between privacy, utility, and scalability. At the operational level, we incorporate **secure MLOps practices**, including encrypted data storage and transmission, strong key management, fine-grained access controls, model provenance tracking, and immutable audit trails, all situated within the shared responsibility model of cloud computing. Building on this synthesis, we propose a **reference architecture** that unifies technical defenses with governance mechanisms and compliance requirements such as GDPR and HIPAA. We also define an **evaluation methodology** that combines adversarial testing with compliance audits. This framework provides actionable guidance for deploying **secure, privacy-preserving ML systems in the cloud** without sacrificing performance or usability.

Keywords: Cloud Machine Learning, Data Security, Privacy Framework, Membership Inference, Model Inversion, Differential Privacy, Federated Learning, Secure MLOps

DEEP LEARNING APPROACHES FOR SOLVING PARTIAL DIFFERENTIAL EQUATIONS

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ABSTRACT

This study investigates the application of deep learning techniques, particularly convolutional neural networks and physics-informed neural networks (PINNs), for approximating solutions of partial differential equations (PDEs). Traditional numerical techniques such as finite difference, finite volume, and finite element methods often face the curse of dimensionality when solving high-dimensional PDEs, making them computationally expensive and time-intensive. Deep learning approaches, on the other hand, can approximate complex mappings in high-dimensional spaces, offering a scalable alternative. By embedding physical laws in neural architectures, PINNs ensure that solutions adhere to the governing equations while benefiting from the flexibility of machine learning. This research evaluates benchmark PDEs in fluid dynamics, heat transfer, and quantum mechanics, comparing computational efficiency with classical solvers.

Keywords

Partial differential equations, Deep learning, Physics-informed neural networks, High-dimensional problems, Computational mathematics

DESIGN AND ANALYSIS OF BLOCKCHAIN BASED E-HEALTH SYSTEM FOR SECURE AND INTEROPERABLE DATA SHARING

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ABSTRACT

The healthcare industry is undergoing a digital transformation , which leads to unprecedented growth in electronics health records(EHRs), telemedicine platforms and Internet of Medical Things(IoMT) data streams. By using, traditional centralized healthcare information systems, it faces continuous challenges like data silos, single points of failure, limited interoperability and risks in data breaches. To overcome all these challenges, this presents the design and analysis of a blockchain-based e-health system that ensures security, interoperability and patient-centric medical data sharing.

This system uses hybrid blockchain architecture that combines public and private blockchains which integrates smart contracts, proxy re-encryption and off-chain storage to enable decentralized and fine-grained access control to ensure data privacy from unauthorized access. In this setup, technology like interplanetary file system(IPFS) a decentralized file storage system used for storing EHRs securely. Smart contracts enforce role and attribute based access control, ensuring that only authorized individuals such as doctors, labs, insurers or researchers can view or modify patient details.

Delegated Proof of Stake (DPos) is a consensus mechanism for efficient and scalable blockchain operation which enhance scalability and reduce computational overhead. HL7FHIR stands for data exchange, which allows the system to work with different hospital and healthcare platform to achieve the interoperability. The security analysis shows the system is well-protected against unauthorized access, replay attack and data tampering. Performance testing further demonstrates impressive gain which is 40% faster in response time and 50% better in scalability over traditional centralized model.

This paper includes advanced privacy techniques such as homomorphic encryption and zero-knowledge proof which analyses the data without exposing the patient information. This study highlights key challenges like compliance with HIPAA and GDPR user onboarding and integrating with existing infrastructure. To tackle these challenges, it focus on AI-powered access control, cross-chain interoperability for multi-institution support and automated compliance auditing using smart contracts. This work offers a detailed blueprint of block-chain based e-health system which leads to a significant improvement in data management, data sharing and data protection to better care coordination and faster innovation across the digital health ecosystem.

Keywords: Smart Contracts, Proxy Re-Encryption, Data Privacy and Security, HL7 FHIR, Internet of Medical Things, Blockchain in Healthcare.

DIGITAL CURRENCIES AND GLOBAL TRADE: THE ROLE OF CRYPTO CURRENCIES AND CBDCS

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ABSTRACT

Digital currencies, including crypto currencies like Bitcoin and Ethereum, and Central Bank Digital Currencies (CBDCs), are rapidly transforming the global financial landscape. Traditionally, cross-border trade has relied on conventional banking systems, which can be slow, costly, and subject to currency volatility. Digital currencies offer faster, more transparent transactions and have the potential to reduce transaction costs while enhancing financial inclusion. CBDCs, issued and regulated by central banks, provide a stable and government-backed alternative to private cryptocurrencies, bridging the gap between traditional fiat money and digital assets. This paper examines how these digital innovations are reshaping international trade, monetary policies, and the future of global finance, exploring both opportunities and challenges in their widespread adoption.

GOLD AND GEOPOLITICAL STABILITY: FUTURE ROLES IN TRADE, INFLATION, AND RESERVES

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ABSTRACT

Gold has historically been a cornerstone of economic security, serving as a universal store of value across civilizations. In the evolving landscape of global finance, its role remains significant, particularly in the context of geopolitical stability. As nations grapple with currency fluctuations, rising inflation, and uncertain trade dynamics, gold offers resilience against systemic risks. This paper explores how gold will continue to influence future global trade, inflation control, and reserve management. It examines gold's function as a hedge against currency volatility, its strategic importance in central bank reserves, and its role in mitigating the economic impact of geopolitical conflicts. Furthermore, the study highlights the balance between traditional reliance on gold and the pressures of modern innovation, such as digital assets and blockchain-based financial systems. By bridging these dimensions, the paper underscores gold's enduring relevance as both a stabilizing force and a strategic tool in the future economy.

DIGITAL PHOTOGRAMMETRY – AN IMAGE ACQUISITION

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ABSTRACT

With the advent of computing and imaging technology, photogrammetry has evolved from analogue to analytical to digital photogrammetry. Digital photogrammetry involves the processing of imagery of all types, including passive (e.g., optical sensing) and active (e.g., radar imaging), taken from any platform (e.g., airborne, satellite, close range, etc.). The unique advantages of digital photogrammetry in terms of precision and accuracy offer opportunities for the automation of Digital Elevation Model (DEM), Digital Terrain Model (DTM), and Digital Surface Model (DSM) generation, as well as the integration of images acquired on a multi-platform and multi-sensor basis. In the field of ancient cultural heritage, 3D accurate models of archaeological objects and sculptures play a significant role in documentation, maintenance, and restoration. Photomapping services provide solutions to the highly specialized needs of today's world, compiling datasets from photogrammetry for land surveying, topographic and geological mapping, engineering design, natural resource management, and environmental inventory. This work aims to highlight the performance of digitization and advancements in calibration technology. The development of cost-effective and accurate tools has made close-range photogrammetry a widely used technique for recovering 3D models of objects with high precision.

Keywords: Digital Photogrammetry, Structure-from-Motion, 3D Surveying, Digital Reconstruction, Mapping Structure, Infrastructure Digitization

DIMENSIONALITY REDUCTION FOR BIG DATA ANALYTICS TO NEGATE TEDIOUS PATTERNS AND ENSURE EFFECTIVE PROCESSING

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ABSTRACT

To evaluate a novel method in Big Data Analytics for Dimensionality Reduction using Deep Association Rule Mining with Deep Learning and Neural Networks to identify complex, non-linear patterns in large datasets. The process involves data pre-processing, creating embedding's, training a deep learning model, extracting, filtering, and selecting association rules, and finally using these rules to reduce data dimensionality, either by removing or consolidating variables. This approach aims to improve the efficiency and interpretability of big data analysis.

Keywords:

Big Data Analytics, Dimensionality Reduction, Pre-processing, Embedding, Association Rule Extraction and Deep Learning Model Training

ENHANCED EMOTION RECOGNITION FOR WOMEN AND CHILDREN SAFETY PREDICTION USING DEEP NETWORK

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ABSTRACT

The safety and security of vulnerable populations, including women and children, in public spaces is a key issue. The ability to detect emotions in real-time which signify, an individual may be in distress will promote timely intervention thereby reducing dangers of harassment or violence. This paper presents a new deep learning emotion recognition framework to detect distress-related emotion by using a multimodal method that combines facial expressions, voice tone, and physiological signals. A convolutional neural network (CNN) and recurrent neural network (RNN) hybrid architecture is utilized to improve emotion classification accuracy in emotions such as fear, anxiety, and anger. We report experimental results on both benchmark datasets and a custom dataset in the real-world, attaining an accuracy of 89.7% on predicting safety-critical emotional states. The proposed system is a scalable solution utilizing an IoT-enabled architecture that can be used in public spaces providing real-time monitoring and warnings for safety, making significant contributions to public urban safety within smart cities.

key words: Emotion Recognition, Deep Learning,CNN,RNN, Real-time Monitoring, Smart Cities, IoT-enabled Systems.

ENHANCED NIGHT VISION OBJECT IDENTIFICATION USING MACHINE LEARNING

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ABSTRACT

This innovation is crucial for applications such as security, independent frameworks, natural life monitoring, and search-and-rescue operations. Create a machine learning-based framework capable of identifying and classifying object sin low-light or no-light conditions utilizing warm and infrared imaging. The framework utilizes convolution neural systems (CNNs) and state-of-the-art question discovery models like YOLOv8(or its upgraded adaptations)for preparing input from night vision cameras. Strong preparing is guaranteed through preprocessing and increasing a data set of warm and infrared pictures, book keeping for conditions like mist, rain, and halfway occlusion. The frame work handles more deceivability, imaging commotion, and natural inconstancy through progressed highlight extraction and multi modal information combination. Measurements like exactness, review and F1-score approve execution, where as edge computing empowers real-time handling for portable and viable applications.

Keywords: Machine Learning, CNN,YOLOv8,F1-Score

ENHANCING INFERTILITY PREDICTION THROUGH FEATURE SELECTION: A COMPARATIVE STUDY OF ENSEMBLE AND CLASSICAL ML MODELS

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ABSTRACT

Infertility is a significant global health challenge, necessitating reliable predictive models for early detection and effective intervention. In this study, we constructed a machine learning pipeline to predict infertility outcomes based on a structured dataset. The dataset initially contained 66 clinical, demographic, and lifestyle features, from which 24 highly relevant predictors were selected using Recursive Feature Elimination (RFE), Chi-square test, and ANOVA feature selection methods. To ensure high-quality inputs, pre processing included scaling and one-hot encoding. Multiple machine learning algorithms were employed, including Random Forest, XGBoost, CatBoost, Support Vector Machine (SVM), Logistic Regression, Naïve Bayes, and AdaBoost, with hyper parameters optimized through Optuna-based tuning. A stacking ensemble combining Naïve Bayes, SVM, Logistic Regression, and AdaBoost with Logistic Regression as the meta-learner was implemented to evaluate the benefits of classifier combination. Model evaluation was carried out using five-fold stratified cross-validation, ensuring robust and unbiased performance comparison. The findings highlight that effective feature selection combined with advanced ensemble learning techniques can significantly improve infertility prediction. These models have strong potential for integration into clinical decision support systems, aiding healthcare professionals in personalized diagnosis and treatment planning.

Keywords: Infertility Prediction, Feature Selection, Ensemble Learning, Random Forest, CatBoost.

ENHANCING PATIENT CARE PREDICTIONS BY MODELING PATIENT SIMILARITIES WITH GRAPHS

CHANDRA PRABA S

ABSTRACT

Accurate patient risk prediction is vital in healthcare for enabling timely intervention and improving clinical outcomes. Conventional predictive models typically rely on tabular data, which often fail to capture the complex relationships between patients and their clinical features. This study proposes a graph-based modeling framework utilizing Graph Neural Networks (GNNs) to enhance predictive analytics in patient care. In this approach, patients are represented as nodes within a graph, and edges are established based on similarity metrics computed from clinical attributes such as demographics, medical history, laboratory results, and treatment patterns. This constructs a patient similarity network that encodes the intricate relational structure of the patient population. The GNN model applies iterative message passing to aggregate feature information from each patient's neighbors, enabling the capture of both individual clinical data and contextual interactions. This AI-driven technique allows the extraction of latent patterns from graphs, improving the prediction of adverse events such as hospital readmissions and disease progression. The framework was evaluated on real-world clinical datasets, demonstrating superior predictive performance compared to traditional machine learning models that do not consider relational data. The results indicate that integrating graph theory with AI-based GNNs facilitates more accurate risk stratification and personalized predictions. This study highlights the potential of advanced graph modeling techniques to transform predictive healthcare analytics and support informed clinical decision-making for improved patient outcomes.

Keywords: Graph Neural Networks (GNNs), Patient Similarity, Predictive Healthcare, Risk Stratification, Machine Learning.

EXPLORING THE STRENGTHS AND TRADE-OFFS OF GANS AND VAEs: A DEEP DIVE INTO MODERN GENERATIVE AI MODELS

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ABSTRACT

Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) are two leading architectures in generative modeling, each with distinct mechanisms and strengths. GANs employ adversarial training between a generator and a discriminator to produce high-quality, sharp images but often face challenges in training stability. VAEs, on the other hand, use a probabilistic framework with an encoder-decoder structure to generate diverse samples and provide meaningful latent representations, though often at the cost of reduced sharpness in outputs. This comparative analysis highlights their architectural differences, training methodologies, output characteristics, and typical applications. GANs excel in producing realistic visual content, making them suitable for tasks demanding high-fidelity generation, whereas VAEs offer robustness and diversity, making them beneficial for tasks requiring interpretability and variation in generated data. Understanding these trade-offs is essential for selecting the appropriate model in different generative AI scenarios, ensuring an optimal balance between image quality, diversity, and interpretability.

Keywords: Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Generative Modeling, Deep Learning, Artificial Intelligence

FINML: MACHINE LEARNING FOR FINANCIAL FRAUD DETECTION

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ABSTRACT

Financial fraud is a rapidly growing threat to global economies, driven by the widespread digitization of financial transactions and increasingly sophisticated tactics employed by fraudsters. Traditional fraud detection methods, which rely heavily on fixed rules and manual oversight, often fall short when faced with evolving and complex attack patterns. This paper proposes a novel security algorithm that integrates advanced machine learning (ML) models with adaptive anomaly detection techniques, cryptographic data validation, and risk-based transaction scoring to enhance the detection of fraudulent activities. By analyzing large volumes of transactional data in real time, the algorithm is able to identify hidden patterns and predict anomalies with high accuracy. The adaptive nature of the system allows it to continuously learn and adjust to emerging fraud strategies, reducing the number of false positives and improving overall response times. Additionally, cryptographic validation ensures data integrity and security, while risk-based scoring prioritizes transactions based on their likelihood of fraud. Together, these components form a robust framework that strengthens the resilience of financial systems against new and sophisticated fraud attempts. This integrated approach not only improves detection efficiency but also fosters greater trust and transparency, ultimately contributing to safer and more reliable financial services.

.Keywords - Financial fraud detection, machine learning, novel security algorithm, adaptive anomaly detection, cryptographic data validation.

GENERATIVE AI FOR DATA AUGMENTATION: METHODS, APPLICATIONS AND CHALLENGES

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ABSTRACT

The effectiveness of machine learning models depends on the quality and quantity of training data. However, real-world datasets often suffer from limitations such as class imbalance, scarcity of labeled samples, noise, and high collection costs. Data augmentation has emerged as a significant technique addressing data scarcity, imbalance, and overfitting in machine learning. Generative Artificial Intelligence offers an effective mechanisms for generating high-quality, diverse data samples that closely reflect real-world distributions. Models such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models can capture complex data distributions and generate high-quality instances across domains including image, text, and time-series data. By expanding dataset diversity, generative approaches not only mitigate overfitting but also improve model generalization and robustness. This paper examines the role of generative AI in data augmentation, outlines key methodologies and highlights emerging applications and challenges in ensuring data validity.

Keywords:

Data augmentation, Generative AI , Machine Learning, Imbalance datasets, Diffusion models

GENERATIVE AI FOR ENHANCED MACHINE LEARNING (ML) DATA AUGMENTATION

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ABSTRACT

Data augmentation has become an essential technique in machine learning to improve model generalization and robustness, particularly in domains where labeled data is scarce, costly, or sensitive. Traditional augmentation strategies, such as geometric transformations in images or noise injection in audio, are limited in their ability to capture complex data distributions. Recent advancements in Generative Artificial Intelligence (GenAI) offer novel approaches for creating high-quality synthetic data across multiple modalities, including text, image, audio, and tabular datasets. Generative Adversarial Networks (GANs), Diffusion Models, and Large Language Models (LLMs) have demonstrated remarkable success in producing diverse and realistic samples, thereby mitigating data imbalance and enhancing downstream model performance. Tools such as CTGAN, StyleGAN, Stable Diffusion, SpecAugment, and enterprise platforms like Gretel.ai and MostlyAI have been effectively applied in domains ranging from medical imaging and automatic speech recognition to financial fraud detection and agricultural monitoring. This paper reviews the landscape of GenAI-based data augmentation, highlighting state-of-the-art techniques, tools, and applications. Furthermore, it discusses the potential of GenAI to address challenges in privacy-preserving data generation while outlining open research issues, including bias mitigation, scalability, and computational cost. The findings underscore that GenAI-driven augmentation is not merely a supplement to traditional methods but a transformative approach that can significantly enhance machine learning performance across diverse real-world scenarios.

Keywords: *Generative Artificial Intelligence (GenAI), Data Augmentation, Generative Adversarial Networks (GANs), Diffusion Models, Large Language Models (LLMs)*

GENERATIVE AI FOR DATA AUGMENTATION: BALANCING SCALABILITY AND DATA RELIABILITY

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ABSTRACT

Generative Artificial Intelligence (AI) has shown significant potential for **data augmentation**, enabling the creation of synthetic datasets to supplement limited or imbalanced real-world data. While such synthetic data can enhance machine learning (ML) model performance, a critical challenge lies in ensuring its **quality, reliability, and ethical use**. Low-quality or biased synthetic data may degrade model accuracy, introduce unintended biases, and lead to harmful predictions. This research investigates methods to **assess, filter, and improve the quality of AI-generated data**. We propose combining automated evaluation metrics with **human-in-the-loop validation** to ensure that augmented datasets are accurate, diverse, and suitable for training ML models. The study will explore the impact of synthetic data quality on model performance across different data types, including text, images, and audio. By developing robust evaluation frameworks and quality assurance strategies, this research aims to maximize the benefits of generative AI for data augmentation while minimizing risks.

Keywords: Generative AI, Data Augmentation, Machine Learning, Data Quality, Human-in-the-Loop, Evaluation Framework

GENERATIVE AI WITH DATA SCIENCE

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ABSTRACT

Generative Artificial Intelligence (AI) and Data Science have emerged as the twin engines driving innovation in today's data-driven world. The integration of these domains enables automated data generation, creative problem solving, and intelligent decision-making. This paper presents a comprehensive exploration of Generative AI and its application within Data Science, reviewing major AI tools such as ChatGPT, Midjourney, GitHub Copilot, and Jasper. It discusses methodologies, literature findings, ethical concerns, and future directions. The study highlights how Generative AI is reshaping industries, enhancing creativity, and enabling scalable data modeling.

Keywords: Generative AI, Data Science, Machine Learning, Automation, Artificial Intelligence.

HYBRID MACHINE LEARNING MODELS FOR RANSOMWARE DETECTION IN CLOUD DATA STORAGE

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ABSTRACT

Cloud computing has quickly become one of the most important technologies of the modern era, offering businesses and individuals flexible, scalable, and cost-effective solutions for storing and managing data. However, the same qualities that make the cloud attractive have also made it a prime target for cybercriminals. Ransomware, in particular, has emerged as one of the most damaging threats. By encrypting or stealing sensitive information and demanding ransom payments, ransomware causes devastating financial, operational, and reputational harm to its victims. Traditional detection methods, such as signature-based systems, are no longer effective against the rapidly evolving nature of ransomware. Even machine learning techniques, when used in isolation, often fail to strike the right balance between accuracy, adaptability, and real-time performance. To address these challenges, this research proposes a hybrid machine learning approach that combines the strengths of Convolutional Neural Networks (CNN) and Support Vector Machines (SVM). The CNN component is responsible for automatically learning complex behavioural features from system and file operation logs, while the SVM provides precise classification of benign and malicious activities. Our experiments, conducted with real-world ransomware datasets and synthetic cloud data, show that the hybrid CNN-SVM model consistently outperforms standalone approaches. With an overall accuracy of 97.2% and an ROC-AUC score of 0.98, the framework demonstrates strong potential for deployment in real-world cloud environments. By reducing false positives, speeding up detection, and providing scalability, the proposed model offers a practical and effective solution for protecting cloud storage systems against ransomware.

KEYWORDS:

Ransomware Detection; Cloud Data Storage; Hybrid Machine Learning; Convolutional Neural Network (CNN); Support Vector Machine (SVM); Cybersecurity; Real-Time Security; Intrusion Detection Systems (IDS); Federated Learning; False Positives Reduction

IDENTIFYING DISEASE CAUSE AND DIAGNOSIS BASED ON CLASS INSTANCEARRIVAL USING DRIFT TECHNIQUES

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ABSTRACT

Concept-drift techniques have been presented for identification of cause of disease and diagnosis. They are the novel class detection in concept-drifting, filter-based data partitioning, data arrival rate using random forests, sentiment distribution using feature-relation networks and so on. A more systematic concept-drift technique enables one to better understand the process and leaves with researchers for further improvements .Most class instance arrival techniques fail in one essential aspect of arrival data i.e. arrival of a class. So, a class instance arrival distribution method that merges a class detection system into traditional classifiers is enabled where the true labels are detected using automatic detection. This paper presents to evaluate the survey on various concept-drift techniques for identifying the disease cause and diagnosis. A qualitative comparison of the approach is presented reducing error rates, providing distribution accuracy.

Keywords: Concept-drift, data partitioning, random forests, sentiment classification, feature-relation networks, class instance arrival

INTEGRATING MACHINE LEARNING TECHNIQUES IN CROP MANAGEMENT: A FRAMEWORK FOR PREDICTIVE ANALYTICS AND RESOURCE OPTIMIZATION

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ABSTRACT

By leveraging both historical weather trends and real-time data from weather applications, machine learning algorithms can deliver precise guidance to farmers for choosing crops, planning irrigation, managing fertilization, and executing other agricultural tasks according to current conditions. To improve resource utilization and promote permaculture, this approach aims to strengthen agriculture in response to environmental changes.

Numerous techniques are employed, including historical data analysis, collaborative weather data integration, infrastructure development, supervised and unsupervised learning, and collaborative training with continuous improvement through retraining. In addition to enhancing accuracy, optimizing crop selection, improving management control, and mitigating climate change effects, these methods help minimize environmental impact.

The goal of augmented agriculture is predictability, achieved through advanced farming techniques such as water, moisture, and rainfall control. By integrating weather APIs with historical temperature data analysis, this solution provides farmers with precise insights for irrigation and other weather-dependent decisions, supporting both water conservation and permaculture practices.

Keywords: Machine Learning, Collaborative Training, Supervised Machine Learning, Unsupervised Machine Learning, Water Management

INTEGRATING TEMPORAL AND SPATIAL FEATURE LEARNING FOR FINANCIAL FRAUD DETECTION WITH LSTM AND CNN

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ABSTRACT

Fraud detection in finance is a pressing challenge, complicated by evolving fraud patterns, data imbalance, and the need for real-time processing. Traditional rule-based and shallow machine learning models are limited in capturing complex temporal and structural relationships within financial transactions. Deep learning offers a powerful alternative, particularly through Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs). LSTMs are well-suited to model sequential dependencies in transaction histories, while CNNs excel at extracting local feature patterns from structured data representations. This paper presents a hybrid deep learning framework that integrates LSTM and CNN architectures to enhance fraud detection performance. We address challenges of class imbalance, scalability, and interpretability, and evaluate the approach against benchmark datasets. Experimental results indicate that the combined LSTM–CNN model achieves superior detection accuracy and robustness compared to single-model approaches, demonstrating its potential for deployment in modern financial systems.

Keywords:

Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNNs), Deep Learning(DL)

INTEGRATING ARTIFICIAL INTELLIGENCE INTO INTRUSION DETECTION FOR ENHANCED NETWORK THREAT ANALYSIS

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ABSTRACT

In today's digital landscape, the exponential growth of network traffic and sensitive data presents critical challenges to cybersecurity. Traditional signature-based intrusion detection systems (IDS) often fail to detect novel and sophisticated attacks. Artificial Intelligence (AI), including Machine Learning (ML), Deep Learning (DL) and ensemble learning, offers promising solutions for automating and enhancing threat detection. This review presents a taxonomy of AI-based IDS approaches, examines existing literature on algorithms, datasets, and evaluation metrics, and highlights their advantages and limitations. Key challenges such as multi-class attack classification, adversarial robustness and interpretability are discussed. Insights are provided for future research to develop adaptive, robust and interpretable intrusion detection systems.

Keywords: Cybersecurity, Intrusion Detection Systems, Machine Learning, Deep Learning, Ensemble Learning, Network Security

IOT SECURITY FRAMEWORKS FOR SMART CITIES

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ABSTRACT

The rapid growth of smart cities relies heavily on the Internet of Things (IoT), where interconnected devices enable efficient urban management, including transportation, healthcare, energy, and public safety. However, this widespread deployment of IoT introduces significant security and privacy challenges due to heterogeneous devices, limited computational resources, and large-scale data exchange. Traditional security solutions often fail to meet the real-time, scalable, and lightweight requirements of smart city environments. This paper presents an overview of IoT security challenges and proposes a comprehensive security framework designed specifically for smart cities. The framework integrates lightweight cryptographic mechanisms, blockchain-based trust management, and artificial intelligence (AI)-driven anomaly detection to ensure data integrity, confidentiality, and resilience against cyberattacks. Furthermore, the framework emphasizes interoperability, scalability, and compliance with privacy regulations. Simulation and case study analysis demonstrate its effectiveness in mitigating threats such as data tampering, denial-of-service (DoS) attacks, and unauthorized access while maintaining system efficiency. This research contributes to the development of secure, sustainable, and trustworthy IoT ecosystems, thereby strengthening the foundation of future smart city infrastructures.

IOT–CLOUD FRAMEWORKS: FEDERATED LEARNING AND XAI FOR ARCHAEOLOGICAL MONITORING

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ABSTRACT

Archaeological research has been significantly enhanced by the integration of Artificial Intelligence (AI) and remote sensing technologies, which enable large-scale, non-invasive site detection and artifact analysis. Despite these advancements, current approaches are hindered by challenges such as limited data accessibility, the opaque nature of AI decision-making, and ethical risks associated with the disclosure of sensitive cultural heritage information. This study introduces a novel framework that integrates federated learning, explainable AI (XAI), and ethical preservation protocols to overcome these limitations. Federated networks are employed to facilitate global collaboration on multisource data, including satellite imagery, LiDAR scans, and drone-based surveys, while preserving local data ownership and privacy. Explainable AI models are developed to enhance interpretability, providing archaeologists with transparent insights into the features and patterns identified by the system. Furthermore, an ethical sensitivity layer is incorporated to classify archaeological findings, ensuring responsible sharing and safeguarding against potential misuse or looting. By combining technical innovation with cultural responsibility, the proposed framework advances AI-powered archaeology by offering a scalable, transparent, and ethically robust approach to the discovery and preservation of human history.

Keywords: Artificial Intelligence, Remote Sensing, Archaeology, Explainable AI, Federated Learning, Cultural Heritage Preservation

LOGIN GUARD BIOMETRIC AUTHENTICATION

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ABSTRACT

In the modern digital era, the growing reliance on online platforms has amplified the need for secure authentication mechanisms. Traditional password-based systems are increasingly vulnerable to attacks such as phishing, brute force, and credential theft. To address these challenges, this paper proposes Login Guard, a biometric-based authentication framework that leverages unique human physiological and behavioral traits for secure user verification. Unlike conventional methods, biometric authentication provides higher reliability by binding access to inherent user characteristics such as fingerprints, facial recognition, or voice patterns, which are difficult to replicate or share. The proposed system integrates multi-modal biometric verification with encryption techniques to ensure both accuracy and data privacy. This study highlights the architecture, implementation strategies, and security benefits of the Login Guard system while evaluating its performance against common cyber threats. Experimental analysis demonstrates that biometric authentication enhances user convenience and significantly reduces unauthorized access risks, making it a robust solution for next-generation authentication systems.

Keywords

Online Authentication, Biometric Security, Login Guard, User Verification, Fingerprint Recognition, Facial Recognition, Voice Recognition, Encryption, Data Privacy, Cybersecurity

MENTAL HEALTH TRACKER AND COMMUNITY SUPPORT PLATFORM

Authors

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ABSTRACT

Mental health has become an increasingly critical concern in today's fast-paced, digitally driven world. Early detection of stress, anxiety, and depression, combined with timely support, can significantly improve well-being. This paper introduces CyberCare, a mental health tracker and community support platform designed to monitor emotional states, provide personalized insights, and connect users with supportive communities. The system integrates self-assessment tools, mood tracking, and data-driven analytics to identify behavioral patterns, while offering access to peer groups, mental wellness resources, and professional guidance. By combining technology with community engagement, CyberCare not only promotes self-awareness but also reduces the stigma around mental health. Experimental evaluation highlights the platform's effectiveness in improving user engagement, fostering positive coping strategies, and creating a safe digital space for collective well-being.

Keywords: Stress deduction, Emotional monitoring, Digital mental health platform, Cyber care, Community support, Peer support, Mental health awareness, personalized insights, behavioural analytics

MACHINE LEARNING AND FEATURE-SELECTION-ENABLED FRAMEWORK FOR ACCURATE CROP YIELD PREDICTION

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ABSTRACT

Agriculture is pivotal to a nation's economy, serving as a primary source of food, employment, and raw materials. However, persistent challenges such as plant diseases, soil degradation, and water scarcity hinder agricultural productivity. The adoption of emerging technologies—particularly machine learning (ML), a branch of artificial intelligence—can help address these challenges by enabling prediction, classification, and automation in agricultural processes. ML techniques optimize irrigation, fertilization, and crop selection, thereby supporting informed decision-making for food security and sustainable crop management.

This study proposes two robust ML architectures for crop classification and yield prediction, leveraging distinct datasets. For crop classification, a crop recommendation dataset from Kaggle is used, which includes features such as soil pH, temperature, humidity, and nutrient levels. Multiple ML classifiers—including Extra Trees Classifier (ETC), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), K-Nearest Neighbors (KNN), Gaussian Naive Bayes (GNB), and Support Vector Machine (SVM)—are evaluated using K-fold cross-validation and Explainable AI (XAI). Random Forest emerged as the top-performing model, achieving 99.7% accuracy, supported by high precision, recall, F1-score, and a well-structured confusion matrix.

This study proposes two robust machine learning architectures for classification and regression based on distinct datasets. Firstly, we delve into a crop recommendation dataset obtained from Kaggle, consisting of various input attributes such as the pH of the soil, temperature, humidity, and nutrient levels. Leveraging machine learning classification techniques such as Extra Tree Classifier (ETC), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), K-Nearest Neighbour (KNN), Gaussian Naive Bayes (GNB), and Support Vector Machine (SVM), we suggest twenty-two different crops founded on these inputs. Through the use of K-fold cross-validation, Explainable AI (XAI) and feature engineering, we identify the best-performing model, with Random Forest coming out on top scoring an accuracy of 99.7% with precision, recall, F1 score, and confusion matrix.

For yield prediction, wheat production data from Pakistan (1992–2013) sourced from the World Bank and FAO is analyzed. Data gaps are addressed using Multivariate Imputation by Chained Equations (MICE). Yield predictions for 2014–2024 and forecasts for 2025 are generated using regression models. After hyperparameter tuning and K-fold cross-validation, the Support Vector Regressor (SVR) achieved the highest accuracy, with an R^2 score of 99.9%.

To enhance transparency and trust, the study incorporates Explainable AI techniques, including Feature Importance and Local Interpretable Model-Agnostic Explanations (LIME), to explain and validate model predictions. The proposed framework can contribute to increasing agricultural productivity, minimizing risks, improving food security, and supporting environmentally sustainable farming practices.

Keywords: Agricultural planning, Crop recommendation, Crop yield forecasting, Machine learning, Explainable AI (XAI), Food security.

NOVEL HYBRID APPROACH FOR MEDICINAL PLANT DISEASE DETECTION COMBINING F-NET AND ViT

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ABSTRACT

Medicinal Plant diseases highly decrease crop yield and quality, which become a challenge in sustainable agriculture. Early and correct diagnosis is important to facilitate prompt interventions and reduce the economic costs. This paper introduces a unified model of automated Medicinal plant disease detection, which uses preprocessing, segmentation, and deep learning-based classification. Histogram Equalization (HE) is used in the preprocessing stage to increase the use of image contrast to normalize the light, making it easier to see disease symptoms. The Thresholding is an automated procedure used by Otsu to partition diseased and healthy leaf tissues to minimize the background interference. To be classified, a hybrid architecture that combines F-NET and the Vision Transformer (ViT) is created, with F-NET learning local texture and structural information, and ViT learning global relationships over the entire leaf surface. Experimental performance on benchmark datasets proves that the proposed framework is better in accuracy, robustness, and generalization as compared to traditional CNN-based methods. The approach offers a viable approach to real-time monitoring of Medicinal plant diseases to aid precision farming and sustainable crop management.

Keywords: ViT, Histogram Equalization, F-Net, CNN, OtsuThresholding

OPERATIONALIZING IEEE P7000 IN AI-DRIVEN SMART CITY SYSTEMS

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ABSTRACT

Modern cities are increasingly automating governance, transportation, energy, and citizen services using artificial intelligence (AI). As these systems shape public life, ethical design becomes essential. This session examines the ethical challenges and technical solutions involved in deploying AI for smart city decision-making, guided by the IEEE P7000 standard for ethical system design. Key concerns such as data privacy, algorithmic bias, explainability, citizen consent, and accountability will be addressed through the lens of this framework, which emphasizes stakeholder engagement and traceable ethical impact analysis throughout the development lifecycle. On the technical front, we will explore how decision trees and reinforcement learning models are applied in urban planning and resource allocation, and how fairness-aware algorithms can be aligned with ethical constraints. The role of explainable AI techniques such as SHAP in fostering civic trust will be highlighted. India's Digital Personal Data Protection (DPDP) law will be discussed in relation to ethical compliance and citizen rights. Case studies from traffic control, welfare delivery, and predictive policing will illustrate how ethical flaws can erode public trust. Finally, we will explore how participatory design and inclusive datasets, when embedded within the IEEE P7000 lifecycle, can ensure that smart city AI systems remain transparent, equitable, and citizen-centric.

Keywords:

Smart Cities, Artificial Intelligence, IEEE P7000, Ethical Design, Data Privacy, Algorithmic Bias, Explainable AI, Citizen Consent, DPDP Act, Fairness-aware Algorithms

OPTIMIZED CROP PRODUCTION ESTIMATION THROUGH GRADIENT BOOSTING AND MLP TECHNIQUES

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ABSTRACT

Agriculture is an important sector in India and serves as a major source of the nation's economy. It is the foremost factor essential for human survival. Machine Learning (ML) offers a crucial approach for developing real-world and effective solutions to crop yield prediction challenges. Globally, agriculture plays a vital role in enhancing the economic contribution of nations. However, many agricultural fields remain underdeveloped due to the limited deployment of ecosystem control technologies. This study aims to enhance agricultural productivity by improving plant yield prediction. Agriculture involves various types of data such as soil, crop, and weather data. Current systems—such as manual counting, climate-smart pest management, and satellite imagery—often produce inaccurate results. Hence, researchers and scientists are focusing on developing better prediction models. The proposed system applies machine learning models that utilize historical data to train the system using algorithms such as **Multilayer Perceptron (MLP)** and **Support Vector Regression (SVR)** to achieve high accuracy. The machine learning regression approach enables better crop yield prediction and improves overall performance. By considering factors such as temperature, rainfall, season, and area, this model provides more precise yield forecasts. Experimental results demonstrate that machine learning significantly improves crop yield prediction, offering valuable insights for optimizing agricultural practices, resource allocation, and risk management. The model is trained and validated using part of the dataset, while the remaining data are used for testing predictive accuracy. This study underscores the potential of machine learning to revolutionize agriculture by enabling precise and timely predictions, ultimately contributing to increased food security and sustainable farming practices.

Keywords:

Machine Learning, Crop Yield Prediction, Regression, Agriculture, Sustainable Farming, Data Analytics

PESTICIDE RECOMMENDATION SYSTEM USING MACHINE LEARNING ALGORITHMS

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ABSTRACT

In modern agriculture, timely and accurate identification of crop diseases and pests is essential for ensuring high productivity and minimizing losses. Traditional methods of diagnosis rely heavily on visual inspection or expert consultation, which can be time-consuming, error-prone, and inaccessible to farmers in remote areas. This project proposes a web-based pesticide recommendation system that leverages machine learning and computer vision to automate disease detection and pesticide selection. The application allows farmers to upload images of diseased or pest-affected plants via a simple web interface. A pre-trained Convolution Neural Network (CNN) model processes the image, identifies the specific disease or pest, and retrieves appropriate pesticide recommendations from a curated, region- specific database. Additionally, the system can issue real-time alerts through SMS or app notifications for severe infections or outbreak situations. The platform supports multiple crop types, maintains historical disease records, and encourages sustainable agricultural practices by minimizing unnecessary pesticide use. This solution aims to empower farmers with quick, accurate, and accessible tools to improve crop health and yield. This solution also addresses key aspects of the Sustainable Development Goal of Responsible Consumption and Production .

Keywords : Pesticides, Recommendation System, CNN, SDG.

PSEUDOCODE FOR GRAPH COLOURING METHOD IN SUSPECT IDENTIFICATION

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ABSTRACT

In this paper, we present the pseudocode for the graph vertex coloring techniques in identification of the suspects during the investigation of crime problem. Pseudocode is an artificial vernacular that assists the designer to create algorithms. This pseudocode will make the job easy and expedite the process of identifying the suspects during crime investigation. The pseudocode functions as a fundamental key of an algorithm which is used for planning a source code which sketches out the structure of the program before the actual coding takes place.

Keywords: vertex coloring, edge coloring, pseudocode, investigation problem.

PLACEMENT PREDICTOR A CAREER GUIDENCY TOOL

Authors

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ABSTRACT

The transition from academics to professional careers poses significant challenges for students, as making informed decisions about job opportunities requires understanding individual strengths and industry demands. The Placement Predictor is a career guidance tool designed to assist students in identifying their likelihood of securing placements in various companies based on academic performance, skill sets, and extracurricular achievements. Utilizing data-driven techniques and predictive algorithms, the system analysis historical placement records and student profiles to provide personalized recommendations. By offering insights into potential career paths, the tool empowers students to focus on areas of improvement, make strategic career choices, and enhance their employability. This project aims to bridge the gap between academic preparation and industry expectations, facilitating a smoother and more informed transition into the professional world.

Keywords: Placement Predictor, Career guidance tool, Data-Driven techniques, Predictive algorithms, Personalized recommendations, Strategic career guidance

POPULATION HEALTH MANAGEMENT (PHM) USING MACHINE LEARNING TECHNIQUES

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ABSTRACT

Population health management, or PHM, has become a crucial strategy for improving health outcomes and reducing costs by shifting the focus from treatment to proactive and preventive care. Recent advances in machine learning (ML) have enabled powerful methods for evaluating large volumes of diverse health data and drawing practical conclusions. This study looks at the application of machine learning in PHM, emphasizing its potential for prevention, early illness detection, and targeted therapies. By using machine learning (ML) techniques like predictive modeling, risk stratification, and pattern recognition, healthcare professionals can identify at-risk groups, diagnose health issues early, and tailor therapies to meet the needs of individuals and communities. We also discuss how ML-driven approaches can enhance patient involvement, optimize resource allocation, and support evidence-based decision-making. Concerns like data privacy, minimizing bias, and incorporating into healthcare workflows are also discussed. All things considered, ML has the potential to drastically alter PHM into a more accurate, proactive, and long-lasting framework, which would eventually result in healthier populations and more efficient healthcare systems.

PREDICTIVE MODELING OF RARE EVENTS IN CREDIT CARD FRAUD DETECTION USING ENSEMBLE LEARNING TECHNIQUES

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ABSTRACT

Predicting rare events in credit card fraud detection using machine learning models is highly challenging when the dataset is highly imbalanced. This study evaluates four machine learning models, namely Logistic Regression, Decision Tree, Random Forest, and XGBoost, on the Kaggle Credit Card Fraud dataset. This dataset is highly imbalanced with less than 0.2% of transactions labeled as fraudulent. The experimental results highlight important differences in model performance of the machine learning algorithms used in this study. Logistic Regression achieves high recall (0.918) but very low precision (0.051), indicating that while most frauds are detected, many normal transactions are incorrectly flagged. Decision Tree demonstrates a more balanced trade-off by substantially improving precision (0.676) and F1-score (0.700), making it more effective at reducing false alarms compared to Logistic Regression. However, its ROC-AUC score (0.862) is lower, suggesting that the model struggles to consistently rank fraudulent transactions higher than legitimate ones. Ensemble methods, Random Forest and XGBoost, deliver the best balance: Random Forest achieves an F1-score of 0.839 and ROC-AUC of 0.953, while XGBoost slightly outperforms it with an F1-score of 0.859 and ROC-AUC of 0.968. These results demonstrate the advantage of ensemble learning, where combining multiple weak learners reduces variance and improves robustness, allowing Random Forest and XGBoost to better handle the severe class imbalance and outperform single models.

Keywords: Credit Card Fraud Detection, Predictive Modeling, Rare Events, Imbalanced Data, Ensemble Learning, Machine Learning Algorithms.

PREDICTIVE MODELING OF DIABETES MELLITUS: A COMPARATIVE STUDY OF TREE-BASED AND KERNEL-BASED APPROACHES

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ABSTRACT

Early detection of type-2 diabetes mellitus can greatly improve patient outcomes and reduce healthcare costs. Machine learning classifiers such as Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM) are frequently applied in medical diagnostic tasks. This study presents a systematic comparison of DT, RF, and SVM on the Pima Indians Diabetes Dataset (PIMA). Preprocessing steps, evaluation methodology, and comparative results are reported across accuracy, precision, recall, F1-score, ROC-AUC, and training time. Trade-offs in interpretability, generalization, and computational cost are also discussed. The experimental findings indicate that Random Forest provides the best balance between predictive performance and robustness, Decision Tree is the fastest and most interpretable, and Support Vector Machine achieves competitive performance with careful hyperparameter tuning. This work is positioned as a reproducible project in applied medical machine learning, making it particularly suitable for undergraduate researchers and academic conferences.

Keywords: Machine Learning, Diabetes prediction, Decision Tree, Random Forest, Support Vector Machine, Classification,

REAL-TIME DATA ANALYTICS ON STREAMING DATA

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ABSTRACT

Real-time data analytics on streaming data is an emerging technology that enables instant analysis of continuously generated data. In the modern digital era, platforms such as Internet of Things (IoT) devices, financial transactions, social media interactions, and e-commerce systems produce massive volumes of high-velocity data every second. Traditional batch processing methods are inadequate to manage this dynamic and fast-changing data environment. This paper presents the concept, necessity, and significance of real-time data analytics in contemporary digital ecosystems. It explores how advanced tools and frameworks—such as Apache Kafka, Apache Spark Streaming, and Apache Flink—facilitate continuous data ingestion, processing, and analysis with low latency and high scalability. The study also examines critical challenges including data accuracy, latency management, fault tolerance, and security concerns. Despite these challenges, real-time analytics offers immense potential in various domains, including instant fraud detection, real-time patient monitoring, intelligent traffic management, and personalized recommendation systems. The paper concludes that real-time data analytics is a transformative approach that enhances decision-making, operational efficiency, and business intelligence. It is a crucial technology for organizations striving to become data-driven and future-ready.

REAL-TIME FRAUD DETECTION IN E-COMMERCE PAYMENTS: USING DEEP LEARNING FOR ADAPTIVE FRAUD PREVENTION

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ABSTRACT

The increasing use of online shopping has also increased attempts at fraud in the form of sophisticated tactics designed to exploit e-commerce payment systems. Legacy fraud detection programs, often run on rule-based algorithms or simple machine learning models, have difficulty anticipating continuously evolving fraud-related activities. This paper aims to explore deep learning methods, especially recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, for real-time fraud in e-commerce payment transactions. By leveraging sequential data from the payment transaction, including payment history, user behaviour, and time series data, this paper will examine the extent to which deep learning (RNNs and LSTMs) can assist investigators in enabling more adaptive, real-time, and potentially more accurate approaches to payment fraud detection and prevention. Empirical experiments will compare the performance on these machines learning models using the same performance as the fraud detection program, but will weigh trade-offs between false positives and computational performance.

Keywords – LSTM,RNN,E-COMMERCE,MACHINE LEARNING.

REALTIME SPEECH EMOTION DETECTION

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ABSTRACT

A real-time system for Speech Emotion Recognition (SER) is developed, leveraging Python and Machine Learning (ML) to identify human emotions directly from spoken language. The system functions by analyzing acoustic features of the speech signal including changes in frequency and speaking speed to classify the speech into specific emotional categories such as happiness, sadness, anger, fear, and neutral. The core value is its capacity to provide immediate emotional insights with high accuracy. This technology has significant practical implications for enhancing customer service in call centres by training customer needs based on emotional state, and for creating more effective and empathetic interactions with devices. By enabling devices to respond sensitively to user emotions, the system aims to create adaptive and personalized user experiences. Furthermore, by offering instant feedback on emotional states, it directly contributes to aspects of Sustainable Development Goal of Good Health and Well-being, by supporting mental health monitoring and non-intrusive emotional assessment. The article uses advanced Machine Learning models trained on large, labelled speech datasets. The final deliverable is an efficient, scalable, and easy-to-integrate solution for emotional intelligence in technology.

Keywords: Machine Learning, Real Time, Speech Emotion Recognition, SDG.

TRANSFORMER-BASED GENERATIVE MODELS: THE RISE OF LARGE LANGUAGE MODELS (LLMs) IN TEXT GENERATION

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ABSTRACT

Transformer-based architectures have revolutionized **Natural Language Processing (NLP)**, enabling significant advancements in text generation through **Large Language Models (LLMs)**. By leveraging **self-attention mechanisms**, **multi-head attention**, and **positional encodings**, transformers effectively capture long-range dependencies and contextual relationships within text, overcoming the limitations of traditional models. Pre-trained on massive and diverse corpora, LLMs generate coherent, contextually relevant, and human-like text across various applications such as **machine translation**, **summarization**, **question answering**, **dialogue systems**, and **content creation**. Despite their remarkable success, challenges such as high computational costs, data biases, interpretability issues, and ethical concerns persist. Current research focuses on enhancing model efficiency, robustness, and explainability through methods like **model pruning**, **quantization**, and **knowledge distillation**, aiming to make LLMs more accessible and ethically responsible for real-world deployment.

Keywords: Transformer, Large Language Models (LLMs), Natural Language Processing (NLP), Text Generation, Self-Attention, Pre-Training, Contextual Embeddings, Ethical AI

VIDEO FILE ENCRYPTION AND DECRYPTION USING ENHANCED VISUAL DEGRADATION METHOD FOR DEEP LEARNING

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ABSTRACT

Many companies have created proprietary or specialized encryption methods for communication and data acquisition. Due to the increasing prevalence of online fraud and cyber theft, safeguarding sensitive data is becoming increasingly critical. Visual cryptography is a prevalent cryptographic method that conceals visual information, including text, photos, videos, and audio, ensuring that only authorized entities with complete access to the encrypted data may decrypt it and disclose the original content, either collectively or individually. Perform several security assessments, including information entropy, mean square error, temporal complexity, histograms, pixel change ratio, peak signal-to-noise ratios, among others, to evaluate the security and robustness of the proposed approaches. The suggested approach The Enhanced Degradation method analyzes video encryption and encoding techniques, employing various assessment metrics to validate the efficacy of the suggested method for video encryption and decryption.

Keywords: Cryptographic, Video Encryption, Advance Encryption Standard, Enhanced Visual Degradation, Digital Security.